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(54) **FLAT BLANK MADE OF AL-MG ALLOY
FOR FLOW FORMING**

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(57) **ABSTRACT**

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The invention relates to a method of manufacturing a flat blank for flow forming comprising DC Casting a rolling ingot having the composition, in weight % Mg: 2.2-2.6, Cr: 0.16-0.25, Fe: 0.10-0.35, Cu: 0.01-0.09, Ti: 0.005-0.15, Si: up to 0.15, Mn: up to 0.08, Zn: up to 0.10, impurities up to 0.05 each and up to 0.15 total, remainder aluminum; Machining said rolling ingot to a slab with a thickness of at least 10 in; Preheating said slab with minimum 2 hours soak at metal temperature of from 1000° F. to 1100° F.; First hot rolling said slab with a break down mill to an intermediate plate with thickness of at least 1.1 in; Second hot rolling said intermediate plate with a tandem mill, with an exit temperature from 650° F. to 690° F., to a plate with a final thickness from 0.27 to 0.48 in; Cutting said plate to obtain a flat blank for flow forming. The invention also relates to a flat blank obtainable by the method of the invention and to the use of a flat blank according to the invention for flow.

Fig. 1

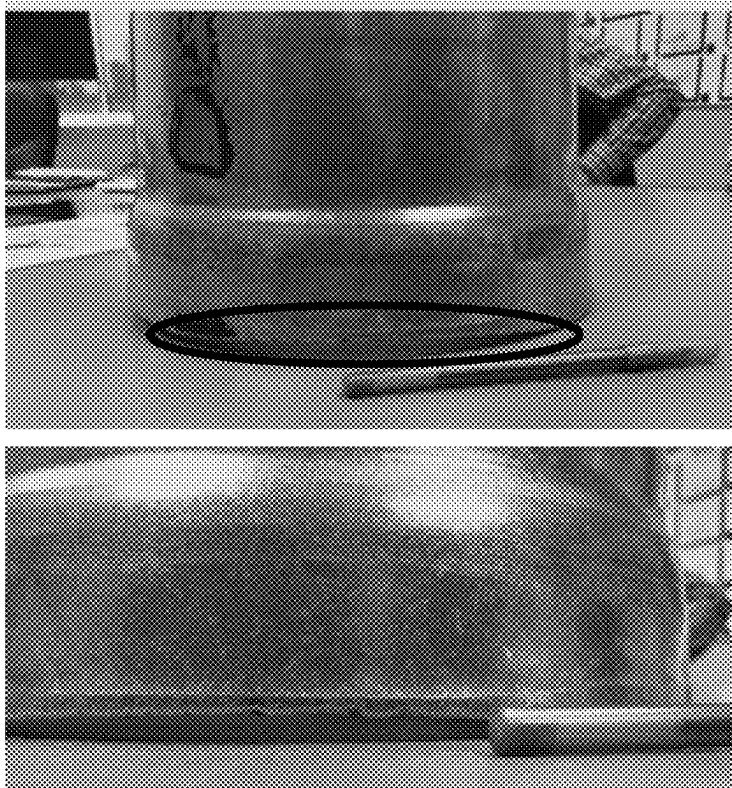


Fig. 2a

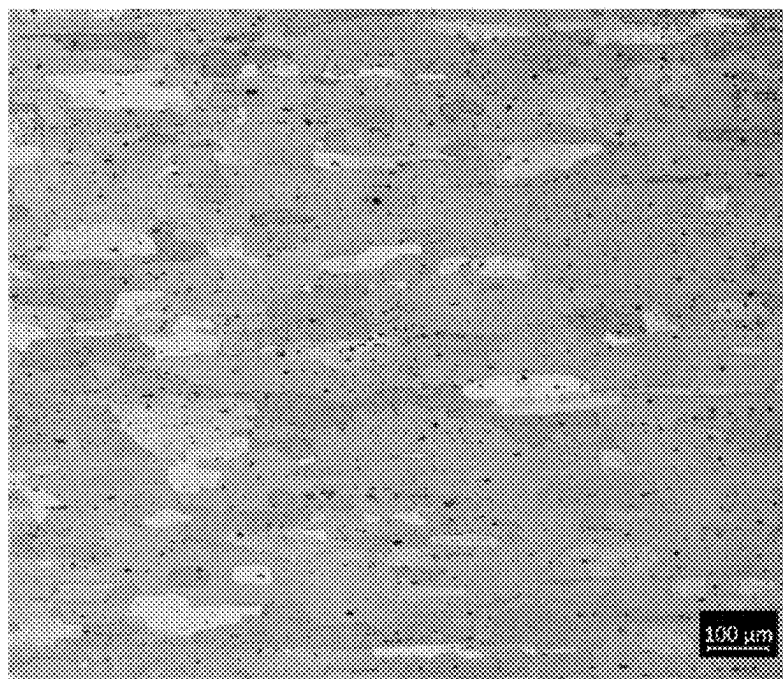
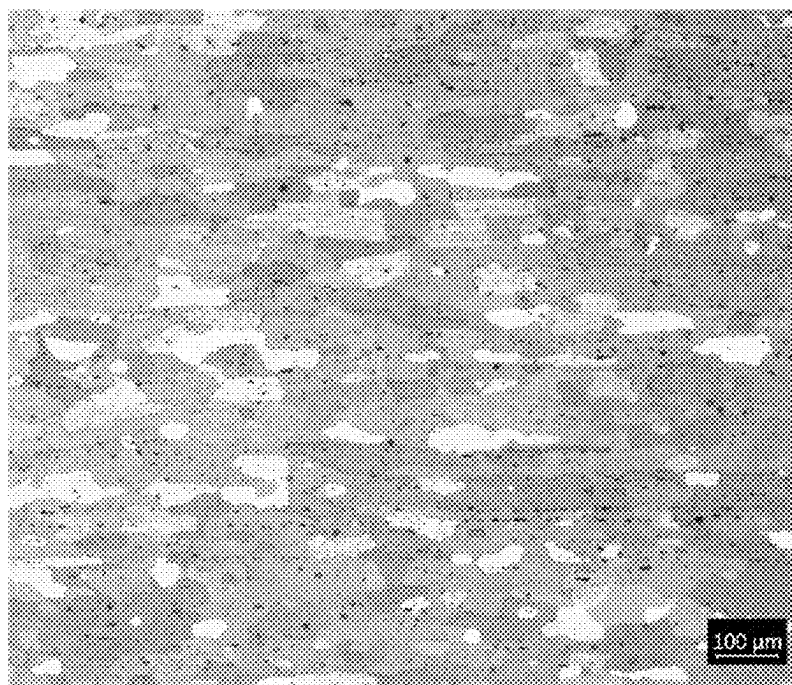


Fig 2b



FLAT BLANK MADE OF AL-MG ALLOY FOR FLOW FORMING

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a non-provisional application claiming priority to U.S. Provisional Application No. 63/557,074, filed on Feb. 23, 2024, the entire contents of which are hereby incorporated by reference in its entirety.

BACKGROUND

Field

[0002] This invention pertains to an aluminum alloy material which is suitable for flat blanking. More particularly, this invention pertains to flat blanks with a thickness typically between 0.27 and 0.48 in made of Al—Mg alloys that are particularly suitable for further flow forming operations towards the production of cylindrical shaped components.

[0003] Spin forming and flow forming are manufacturing processes, typically performed close to ambient temperature, that produce single-piece, axisymmetric parts. Flow forming in particular is a fairly new technique used for the production of dimensionally accurate near net shaped hollow components. Both processes rely on the action of sets of movable rollers on a rotating blank to plastically deform the material. Spin forming begins with a flat preform, which is shaped into curved or cup-shaped articles in free space. The diameter decreases without appreciable changes to the wall thickness. Flow forming typically begins with a cylindrical preform, which is drawn out against a mandrel. The length increases, the wall thickness decreases, and the inner diameter may change. The origins of the flow forming process started in Sweden in the mid-1950s. Flow forming is an advanced form of the metal spinning process and is sometimes called tube spinning as well. In the flow forming process there is the simultaneous extrusion (or drawing) and rolling of the workpiece. Therefore, flow forming is a very complex process comprising many effects of the different operations.

[0004] The flat preform or blank, when it is made of an aluminum alloy is usually a plate with a thickness typically between 0.27 and 0.48 in.

[0005] The technique is useful in several industries such as aerospace as described by Mulvaney et al. (Annual Virginia Space Grant Consortium (VSGC) Student Research Conference, Newport News, VA, Apr. 13, 2023) or for automotive parts.

[0006] Among automotive parts, flow forming is particularly useful to make transmission hubs (also known as transmission houses or shells or case) which are part of the transmission. For example, patent applications WO 02/28561 discloses a method for cold forming splined transmission hubs. Patent application WO2007027992 discloses a method and tool for forming a flow formed part, wherein the tool has an annular array of spline forming recesses for producing a flow formed part having splines. Patent application US2023/039866 discloses a method of manufacturing a torque-transmitting component including providing a flat blank to a transfer press having a plurality of stations and performing a plurality of pressing operations, in which the flat blank is formed into a cup shape, rough splines are formed on the cup shape, and the rough splines are further pressed to define smooth splines.

[0007] The aerospace or automotive parts obtained by the spin forming and flow forming manufacturing processes are

designed to manufacture high precision pieces. The quality of the flat blank that is used for these manufacturing processes is particularly sensitive for parts with long dimensions and/or thick side walls.

[0008] A problem addressed by the present application is to make flat blanks of an aluminum magnesium alloy particularly suited for the flow forming manufacturing processes, in particular flow forming processes which result in components with either thick side walls or a long cylindrical dimension; in either case the flat blank material shall allow for material movement suitable to form the component with minimal rework after the flow forming operation.

SUMMARY

[0009] The invention relates to a method of manufacturing a flat blank for flow forming comprising

[0010] (a) DC Casting a rolling ingot having the composition, in weight %

[0011] Mg: 2.2-2.6,

[0012] Cr: 0.16-0.25,

[0013] Fe: 0.10-0.35,

[0014] Cu: 0.01-0.09,

[0015] Ti: 0.005-0.15,

[0016] Si: up to 0.15,

[0017] Mn: up to 0.08,

[0018] Zn: up to 0.10,

[0019] impurities up to 0.05 each and up to 0.15 total, remainder aluminum,

[0020] (b) Machining said rolling ingot to a slab with a thickness of at least 10 in,

[0021] (c) Preheating said slab with minimum 2 hours soak at metal temperature of from 1000° F. to 1100° F.,

[0022] (d) First hot rolling said slab with a break down mill to an intermediate plate with thickness of at least 1.1 in,

[0023] (e) Second hot rolling said intermediate plate with a tandem mill, with an exit temperature from 650° F. to 690° F., to a plate with a final thickness from 0.27 to 0.48 in,

[0024] (f) Cutting said plate to obtain a flat blank for flow forming.

[0025] The invention also relates to a flat blank obtainable by the method of the invention and to the use of a flat blank according to the invention for flow forming.

ILLUSTRATIONS

[0026] As used below, any reference to a series of illustrations is to be understood as a reference to each of those examples disjunctively (e.g., “Illustrations 1-4” is to be understood as “Illustrations 1, 2, 3, or 4”).

[0027] Illustration 1. A method of manufacturing a flat blank for flow forming comprising

[0028] (a) DC Casting a rolling ingot having the composition, in weight %

[0029] Mg: 2.2-2.6, preferably 2.3 to 2.5,

[0030] Cr: 0.16-0.25, preferably 0.19 to 0.22,

[0031] Fe: 0.10-0.35, preferably 0.20 to 0.30,

[0032] Cu: 0.01-0.09, preferably 0.02 to 0.07,

[0033] Ti: 0.005-0.15, preferably 0.01 to 0.05,

[0034] Si: up to 0.15, preferably 0.05 to 0.13,

[0035] Mn: up to 0.08, preferably 0.03 to 0.07,

[0036] Zn: up to 0.10, preferably up to 0.05,

[0037] impurities up to 0.05 each and up to 0.15 total, remainder aluminum,

- [0038] (b) and machining said rolling ingot to a slab with a thickness of at least 10 in,
- [0039] (c) Preheating said slab with minimum 2 hours soak at metal temperature of from 1000 OF to 1100° F. preferably from 1025° F. to 1075° F.,
- [0040] (d) First hot rolling said slab with a break down mill to an intermediate plate with thickness of at least 1.1 in, preferably from 1.2 in to 1.4 in,
- [0041] (e) Second hot rolling said intermediate plate with a tandem mill, with an exit temperature from 650° F. to 690° F., preferably from 660° F. to 680° F. to a plate with a final thickness from 0.27 to 0.48 in, preferably from 0.31 to 0.43 in,
- [0042] (f) Cutting said plate to obtain a flat blank for flow forming.
- [0043] Illustration 2. A method according to illustration 1 wherein in step (d) the entry temperature is at least 850° F. preferably at least 870° F. and/or is at most 950° F. preferably at most 930° F.
- [0044] Illustration 3. A flat blank for flow forming obtainable by the method of illustration 1.
- [0045] Illustration 4. A flat blank according to illustration 3 wherein the TYS is from 8 to 25 ksi in any direction and wherein the spread of TYS in any direction is at most 12 ksi.
- [0046] Illustration 5. A flat blank according to illustration 3 or illustration 4 wherein the UTS is from 25 to 30 ksi in any direction and wherein the spread of UTS in any direction is at most 10 ksi.
- [0047] Illustration 6. A flat blank according to any one of illustrations 3 to 5 wherein the grain size measured at mid-thickness on a L/ST section in the short direction is at least 40 μm , preferably at least 43 μm .
- [0048] Illustration 7. A flat blank according to any one of illustrations 3 to 6 wherein the difference between hardening texture, quantified by the sum of the bulk volume fraction of the “copper”, “brass” and “S” orientations and softening texture, quantified by the sum of the bulk volume fraction of “Cube”, “Goss” and “Rotated cube” orientations is no larger than 7%, preferably no larger 6%, even more preferably, no larger than 5%.
- [0049] Illustration 8. Use of a flat blank according to any one of illustrations 3 to 7 for flow forming.
- [0050] Illustration 9. Use of a flat blank according to illustration 8 for manufacturing by flow forming a cylindrical part that has side walls with a thickness of at least 0.08 in preferably at least 0.09 in and/or a cylindrical dimension with a length of at least 8 in preferably at least 9 in.
- [0051] Illustration 10. Use according to illustration 8 or illustration 9 wherein the cylindrical part is used to make a transmission housing.

BRIEF DESCRIPTION OF THE DRAWINGS

- [0052] FIG. 1 illustrates a flow formed cylindrical part that has side walls with a thickness of about 0.1 in and/a length of at about 10 in, with an earing behavior that is not within the invention.
- [0053] FIG. 2 illustrates the grain size measurement at mid-thickness on a L/ST section for example A (FIG. 2a) and B (FIG. 2b).

DETAILED DESCRIPTION OF A PREFERRED EMBODIMENT

[0054] Unless otherwise indicated, all indications concerning the chemical composition of alloys are expressed as a percentage by weight based on the total weight of the alloy. The expression 1.4 Cu means that the copper content expressed in % by weight is multiplied by 1.4. The designation of the alloys is made in accordance with the regulations of The Aluminum Association, known to those skilled in the art.

[0055] The static mechanical properties in tension, in other words the ultimate tensile strength UTS, the conventional tensile yield strength at 0.2% elongation TYS, and the elongation at break A %, are determined by a tensile test according to standard ASTM E8 and ASTM B557.

[0056] Unless stated otherwise, the definitions of standard EN 12258 (2012) apply.

[0057] The crystallographic texture can be described by a 3-dimensional mathematical function. This function is known in the trade as the Orientation Density Function (ODF). It is defined as the volume fraction of material dV/V having a $g \pm dg$ orientation:

$$\frac{dV/V}{dg} = f(g) = f(\varphi_1, \Phi, \varphi_2)$$

[0058] where (ϕ_1, Φ, ϕ_2) are the Euler angles describing the orientation g .

[0059] The present inventors have calculated the ODF of each plate using the method of spherical harmonics from four pole figures measured by X-ray diffraction on a conventional texture goniometer. The ATEX© software was used for analysis. The pole figure measurements were performed on samples cut through-thickness and stacked up to make a sandwich 50 mm short-transverse direction and 20 mm in transverse longitudinal direction. This testing allowed for the average texture of the complete thickness, referred to herein as bulk texture, of the plate to be measured and analyzed.

[0060] It is usual to simplify the information in the ODF by calculating the volume fraction of crystallites that have a specific orientation. To do this, one arbitrarily defines the reference orientation and an angle of maximum disorientation around this orientation. The ODF is then integrated into the domain so defined, which makes it possible to deduce the relative volume of orientations in this domain relative to the total volume. The present inventors used a tolerance of 15° around “copper”, “brass” and “S” orientation, the “hardening textures” and Cube, Goss, Rotated cube ($\text{CG}_{26.5}$) orientations, the “softening textures” in order to describe the texture obtained. Crystallographic orientations are known to those skilled in the art and are described for example in the reference document by U. F. Kocks, C. N. Tomé and H.-R. Wenk, “Texture and anisotropy: preferred orientations in polycrystals and their effect on materials properties”. Cambridge University Press, 2000. The orientations are given in the table below.

Name	Indices	Bunge (ϕ_1 , Φ , ϕ_2)
Copper	{112}<111>	90, 35, 45
S	{123}<634>	59, 37, 63
Brass	{110}<112>	35, 45, 0
Cube	{001}<100>	0, 0, 0
Goss	{011}<100>	0, 45, 0
CG _{26, 5}	{021}<100>	0, 26, 0

[0061] According to the method of the invention, a rolling ingot is cast by DC casting, or Direct Chill casting which is a semi-continuous casting process, and machined to obtain a slab with a thickness of at least 10 in. Preferably the slab thickness is at least 15 or 19 or 20 or 21 in. The machining step, also known as scalping, enables to withdraw typically at least about 0.5 in on each large face of the rolling ingot and provides an optimized surface quality for rolling, which also affects the flat blank surface quality.

[0062] The composition is, in weight %, Mg: 2.2-2.6, preferably 2.3 to 2.5, Cr: 0.16-0.25, preferably 0.19 to 0.22, Fe: 0.10-0.35, preferably 0.20 to 0.30, Cu: 0.01-0.09, preferably 0.02 to 0.07, Ti: 0.005-0.15, preferably 0.01 to 0.05, Si: up to 0.15, preferably 0.05 to 0.13, Mn: up to 0.08, preferably 0.03 to 0.07, Zn: up to 0.10, preferably up to 0.05, impurities up to 0.05 each and up to 0.15 total, remainder aluminum.

[0063] In an embodiment, the content of Mg is at least about 2.2%, and at most about 2.3% or at most about 2.4% or at most about 2.5% or at most about 2.6%. In an embodiment, the content of Mg is at least about 2.3%, and at most about 2.4% or at most about 2.5% or at most about 2.6%. In an embodiment, the content of Mg is at least about 2.4%, and at most about 2.5% or at most about 2.6%. In an embodiment, the content of Mg is at least about 2.5%, and at most about 2.6%. Preferably the Mg content is from 2.3 to 2.5.

[0064] In an embodiment, the content of Cr is at least about 0.16%, and at most about 0.18% or at most about 0.20% or at most about 0.22% or at most about 0.24%. In an embodiment, the content of Cr is at least about 0.18%, and at most about 0.20% or at most about 0.22% or at most about 0.24%. In an embodiment, the content of Cr is at least about 0.20%, and at most about 0.22% or at most about 0.24%. In an embodiment, the content of Cr is at least about 0.22%, and at most about 0.24%. Preferably the Cr content is from 0.19 to 0.22.

[0065] In an embodiment, the content of Fe is at least about 0.10%, and at most about 0.13% or at most about 0.16% or at most about 0.19% or at most about 0.22% or at most about 0.25% or at most about 0.28% or at most about 0.31% or at most about 0.34%. In an embodiment, the content of Fe is at least about 0.13%, and at most about 0.16% or at most about 0.19% or at most about 0.22% or at most about 0.25% or at most about 0.28% or at most about 0.31% or at most about 0.34%. In an embodiment, the content of Fe is at least about 0.16%, and at most about 0.19% or at most about 0.22% or at most about 0.25% or at most about 0.28% or at most about 0.31% or at most about 0.34%. In an embodiment, the content of Fe is at least about 0.19%, and at most about 0.22% or at most about 0.25% or at most about 0.28% or at most about 0.31% or at most about 0.34%. In an embodiment, the content of Fe is at least about 0.22%, and at most about 0.25% or at most about 0.28% or at most about 0.31% or at most about 0.34%. In an embodiment, the content of Fe is at least about 0.25%, and at most about 0.28% or at most about 0.31% or at most about 0.34%. In an embodiment, the content of Fe is at least about 0.28%,

and at most about 0.31% or at most about 0.34%. In an embodiment, the content of Fe is at least about 0.31%, and at most about 0.34%. Preferably the Fe content is from 0.20 to 0.30.

[0066] In an embodiment, the content of Cu is at least about 0.01%, and at most about 0.02% or at most about 0.03% or at most about 0.04% or at most about 0.05% or at most about 0.06% or at most about 0.07% or at most about 0.08% or at most about 0.09%. In an embodiment, the content of Cu is at least about 0.02%, and at most about 0.03% or at most about 0.04% or at most about 0.05% or at most about 0.06% or at most about 0.07% or at most about 0.08% or at most about 0.09%. In an embodiment, the content of Cu is at least about 0.03%, and at most about 0.04% or at most about 0.05% or at most about 0.06% or at most about 0.07% or at most about 0.08% or at most about 0.09%. In an embodiment, the content of Cu is at least about 0.04%, and at most about 0.05% or at most about 0.06% or at most about 0.07% or at most about 0.08% or at most about 0.09%. In an embodiment, the content of Cu is at least about 0.05%, and at most about 0.06% or at most about 0.07% or at most about 0.08% or at most about 0.09%. In an embodiment, the content of Cu is at least about 0.06%, and at most about 0.07% or at most about 0.08% or at most about 0.09%. In an embodiment, the content of Cu is at least about 0.07%, and at most about 0.08% or at most about 0.09%. In an embodiment, the content of Cu is at least about 0.08%, and at most about 0.09%. Preferably the Cu content is from 0.02 to 0.07.

[0067] In an embodiment, the content of Ti is at least about 0.01%, and at most about 0.03% or at most about 0.05% or at most about 0.07% or at most about 0.09% or at most about 0.11% or at most about 0.13% or at most about 0.15%. In an embodiment, the content of Ti is at least about 0.03%, and at most about 0.05% or at most about 0.07% or at most about 0.09% or at most about 0.11% or at most about 0.13% or at most about 0.15%. In an embodiment, the content of Ti is at least about 0.05%, and at most about 0.07% or at most about 0.09% or at most about 0.11% or at most about 0.13% or at most about 0.15%. In an embodiment, the content of Ti is at least about 0.07%, and at most about 0.09% or at most about 0.11% or at most about 0.13% or at most about 0.15%. In an embodiment, the content of Ti is at least about 0.09%, and at most about 0.11% or at most about 0.13% or at most about 0.15%. In an embodiment, the content of Ti is at least about 0.11%, and at most about 0.13% or at most about 0.15%. In an embodiment, the content of Ti is at least about 0.13%, and at most about 0.15%. Preferably the Ti content is from 0.01 to 0.05.

[0068] In an embodiment, the content of Si is at most about 0.02% or at most about 0.04% or at most about 0.06% or at most about 0.08% or at most about 0.10% or at most about 0.12% or at most about 0.14%. In an embodiment, the content of Si is at least about 0.02%, and at most about 0.04% or at most about 0.06% or at most about 0.08% or at most about 0.10% or at most about 0.12% or at most about 0.14%. In an embodiment, the content of Si is at least about 0.04%, and at most about 0.06% or at most about 0.08% or at most about 0.10% or at most about 0.12% or at most about 0.14%. In an embodiment, the content of Si is at least about 0.06%, and at most about 0.08% or at most about 0.10% or at most about 0.12% or at most about 0.14%. In an embodiment, the content of Si is at least about 0.08%, and at most about 0.10% or at most about 0.12% or at most about 0.14%. In an embodiment, the content of Si is at least about 0.10%, and at most about 0.12% or at most about 0.14%. In an embodiment, the content of Si is at least about 0.12%, and at most about 0.14%.

embodiment, the content of Si is at least about 0.12%, and at most about 0.14%. Preferably the Si content is from 0.05 to 0.13.

[0069] In an embodiment, the content of Mn is at most about 0.01% or at most about 0.02% or at most about 0.03% or at most about 0.04% or at most about 0.05% or at most about 0.06% or at most about 0.07% or at most about 0.08%. In an embodiment, the content of Mn is at least about 0.01%, and at most about 0.02% or at most about 0.03% or at most about 0.04% or at most about 0.05% or at most about 0.06% or at most about 0.07% or at most about 0.08%. In an embodiment, the content of Mn is at least about 0.02%, and at most about 0.03% or at most about 0.04% or at most about 0.05% or at most about 0.06% or at most about 0.07% or at most about 0.08%. In an embodiment, the content of Mn is at least about 0.03%, and at most about 0.04% or at most about 0.05% or at most about 0.06% or at most about 0.07% or at most about 0.08%. In an embodiment, the content of Mn is at least about 0.04%, and at most about 0.05% or at most about 0.06% or at most about 0.07% or at most about 0.08%. In an embodiment, the content of Mn is at least about 0.05%, and at most about 0.06% or at most about 0.07% or at most about 0.08%. In an embodiment, the content of Mn is at least about 0.06%, and at most about 0.07% or at most about 0.08%. In an embodiment, the content of Mn is at least about 0.07%, and at most about 0.08%. Preferably the Mn content is from 0.03 to 0.07.

[0070] In an embodiment, the content of Zn is at most about 0.01% or at most about 0.02% or at most about 0.03% or at most about 0.04% or at most about 0.05% or at most about 0.06% or at most about 0.07% or at most about 0.08% or at most about 0.09% or at most about 0.10%. In an embodiment, the content of Zn is at least about 0.01%, and at most about 0.02% or at most about 0.03% or at most about 0.04% or at most about 0.05% or at most about 0.06% or at most about 0.07% or at most about 0.08% or at most about 0.09% or at most about 0.10%. In an embodiment, the content of Zn is at least about 0.02%, and at most about 0.03% or at most about 0.04% or at most about 0.05% or at most about 0.06% or at most about 0.07% or at most about 0.08% or at most about 0.09% or at most about 0.10%. In an embodiment, the content of Zn is at least about 0.03%, and at most about 0.04% or at most about 0.05% or at most about 0.06% or at most about 0.07% or at most about 0.08% or at most about 0.09% or at most about 0.10%. In an embodiment, the content of Zn is at least about 0.04%, and at most about 0.05% or at most about 0.06% or at most about 0.07% or at most about 0.08% or at most about 0.09% or at most about 0.10%. In an embodiment, the content of Zn is at least about 0.05%, and at most about 0.06% or at most about 0.07% or at most about 0.08% or at most about 0.09% or at most about 0.10%. In an embodiment, the content of Zn is at least about 0.06%, and at most about 0.07% or at most about 0.08% or at most about 0.09% or at most about 0.10%. In an embodiment, the content of Zn is at least about 0.07%, and at most about 0.08% or at most about 0.09% or at most about 0.10%. In an embodiment, the content of Zn is at least about 0.08%, and at most about 0.09% or at most about 0.10%. In an embodiment, the content of Zn is at least about 0.09%, and at most about 0.10%. Preferably the zinc content is up to 0.05.

[0071] Impurities are up to 0.05 each and up to 0.15 total, remainder aluminum. Impurities are elements that are not intentionally added or incidental to the addition of other elements.

[0072] The slab is preheated with minimum 2 hours soak at metal temperature of from 1000° F. to 1100° F. preferably

from 1025° F. to 1075° F. Preheating the slab is useful to reach the desired metallurgical properties such as grain size and texture.

[0073] A first hot rolling said rolling ingot with a break down mill to an intermediate plate with thickness of at least 1.1 in, preferably from 1.2 in to 1.4 in, the entry temperature being in an embodiment at least 850° F. preferably at least 870° F. and/or at most 950° F. preferably at most 930° F. A second hot rolling of said intermediate plate with a tandem mill is then carried out, with an exit temperature from 650° F. to 690° F., preferably from 660° F. to 680° F. to a plate with a final thickness from 0.27 to 0.48 in, preferably from 0.31 to 0.43 in. The tandem mill may be typically a three stands or a four stands or a five stands tandem mill. Control of the hot rolling temperatures and thicknesses is also useful to reach the desired metallurgical properties such as grain size and texture.

[0074] The plate at final thickness is then cut into blanks suitable for flow forming.

[0075] The present inventors have found that by combining specific composition of a specific Al—Mg alloy and process steps the problem is solved and flat blanks particularly suitable for flow forming are obtained.

[0076] In particular, by combining a specifically controlled composition and process conditions, the present inventors were able to obtain flat blanks having a low anisotropy. According to the present inventors, the improved properties may be related to an improved control of the recrystallization and a more restored microstructure during and following hot rolling. The flat blanks according to the invention are found to have larger grain sizes through the thickness than reference materials, which also have a grain alignment relative to the rolling direction more pronounced than those according to the invention. A higher amount of large-shaped grains and overall higher isotropic grain alignment was found to lead to improved formability under flow forming operations. In an embodiment, the grain size measured at mid-thickness on a L/ST section in the short direction is at least 40 μm , preferably at least 43 μm .

[0077] In an embodiment, the TYS of the flat blank of the invention is from 8 to 25 ksi, preferably 9 to 22 ksi in any direction and the spread of TYS in any direction is at most 12 ksi, preferably at most 10 ksi.

[0078] In an embodiment the UTS is from 25 to 30 ksi preferably from 26 to 29 ksi in any direction and the spread of UTS in any direction is at most 10 ksi, preferably at most 5 ksi.

[0079] In an embodiment, the elongation is at least 18% in the L direction, preferably at least 21% and more preferably at least 23%.

[0080] The present inventors also observed specific texture features for the product according to the invention. The products according to the invention exhibit a low difference between the total of these two groups of textures, softening and hardening. According to the inventors, this could be related to an observed reduction in the total earing. Indeed, as a result of different radial elongations in different directions of the flat blank during flow forming, undesired wavy rims form, which is called earing. The highest parts are called ear peaks, while the lowest regions are called ear valleys. Earing formation, which is related to texture and mechanical properties, results from a complex combination between composition and processing. A uniform texture distribution seems to improve the flow formability by equal distribution of strain deformation (plasticity) during flow forming operations and helps to reduce the undesirable large earing at the end of the part formation. FIG. 1 illustrates a

flow formed part having a poor earing behavior. In an embodiment, for the flat blank according to the invention the difference between hardening texture, quantified by the sum of the bulk volume fraction of the “copper”, “brass” and “S” orientations and softening texture, quantified by the sum of the bulk volume fraction of “Cube”, “Goss” and “Rotated

with an entry temperature of about 870° F. and an exit temperature of about 550° F. The plates were cut into flat blanks.

[0084] The mechanical properties of flat blanks A and B were characterized in three different directions. The results are presented in Table 2

TABLE 2

Mechanical properties in ksi (TYS and UTS) and % (A %) in the Longitudinal (L), 45° and Transverse (LT) directions											
	TYS- L	UTS- L	A %- L	TYS- 45	UTS- 45	A %- 45	TYS- LT	UTS- LT	A %- LT	TYS (max – min)	UTS (max – min)
A	19.4	28.7	24.0	10.6	27.0	33.5	12.1	27.8	30.2	8.8	1.7
B	14.3	28.3	27.0	13.0	27.2	33.0	14.0	28.0	27.9	14.3	28.3

cube” orientations is no larger than 7%, preferably no larger 6%, even more preferably, no larger than 5%. By bulk volume fraction it is meant average volume fraction throughout the full thickness of the flat blank.

[0081] The flat blanks according to the invention are useful for flow forming, in particular for manufacturing cylindrical parts that have thick side walls with a thickness of at least 0.08 in preferably at least 0.09 in and/or a long cylindrical dimension with a length of at least 8 in preferably at least 9 in. The cylindrical part is particularly used to make a transmission housing.

[0085] The mechanical properties are much more isotropic for the flat blanks A than for plate B.

[0086] The grain size, measured at mid-thickness on a L/ST section in the short direction at ×100 magnification is illustrated by FIG. 2a, product A and 2b, product B. For product A the grain size was 47 μm whereas for product B it was 35 μm.

[0087] The texture of the flat blanks A and B were measured, the results, in % of volume fraction, are reported in Table 3

TABLE 3

bulk volume fraction (%) of the texture orientations									
	Copper	Brass	S	Hardening Texture	Cube	Goss	Rotated Cube	Softening Texture	Softening – Hardening
A	2.0	9.5	10.5	22.0	14.7	2.1	8.8	25.6	3.6
B	1.4	6.8	7.5	15.7	14.2	2.0	7.9	24.0	8.3

EXAMPLES

Example 1

[0082] An aluminum alloy with a composition according to Table 1 was cast and into two a 24 in thick rolling ingots and machined to obtain two slabs 23 in thick.

TABLE 1

Composition in wt. %								
	Si	Fe	Cu	Mn	Mg	Cr	Zn	Ti
A&B	0.11	0.28	0.04	0.06	2.45	0.19	0.03	0.01

[0083] The slabs were pre-heated with minimum 2 hours soak at metal temperature of 1025-1075 F The slabs were hot rolled on a break down mill to intermediate plates having a thickness of 1.3 in with a lay on temperature of about 900° F. A first intermediate plate A was further hot rolled into a plate 0.34 in thick with a 5 stands tandem mill with an entry temperature of about 870° F. and an exit temperature of about 670° F. A second intermediate plate B was further hot rolled into a plate 0.34 in thick with a 5 stands tandem mill

[0088] For the flat blank A according to the invention the difference between hardening texture, quantified by the sum of the bulk volume fraction of the “copper”, “brass” and “S” orientations and softening texture, quantified by the sum of the bulk volume fraction of “Cube”, “Goss” and “Rotated cube” orientations less than 4% whereas for the flat blank B it was more than 8%.

Example 2

[0089] Aluminum alloys with a composition according to Table 4 were cast into two a 24 in thick rolling ingots and machined into two slabs 23 in thick. The chromium content of ingot D is not within the scope of the invention.

TABLE 4

Composition in wt. %								
	Si	Fe	Cu	Mn	Mg	Cr	Zn	Ti
C	0.09	0.23	0.04	0.07	2.39	0.22	0.03	0.01
D	0.09	0.24	0.05	0.06	2.39	0.26	0.06	0.01

[0090] The slabs were pre-heated with minimum 2 hours soak at metal temperature of 1025-1075° F. The slabs were

then hot rolled on a break down mill to intermediate plates having a thickness of 1.3 in with a lay on temperature of about 891° F. for alloy C and 897° F. for alloy D. The intermediate plate made of alloy C was further hot rolled into a plate 0.34 in thick with a 5 stands tandem mill with an entry temperature of about 867° F. and an exit temperature of about 669° F. The intermediate plate made of alloy D was further hot rolled into a plate 0.34 in thick with a 5 stands tandem mill with an entry temperature of about 857° F. and an exit temperature of about 669° F.

[0091] The plates C and D were flow formed. Plate C provided good results whereas for plate D earing occurred as shown in FIG. 1. This amount of earing requires reworking of the piece and is not considered a successfully flow formed part.

1. A method of manufacturing a flat blank for flow forming comprising

(a) DC Casting a rolling ingot having a composition, in weight %

Mg: 2.2-2.6, optionally 2.3 to 2.5,

Cr: 0.16-0.25, optionally 0.19 to 0.22,

Fe: 0.10-0.35, optionally 0.20 to 0.30,

Cu: 0.01-0.09, optionally 0.02 to 0.07,

Ti: 0.005-0.15, optionally 0.01 to 0.05,

Si: up to 0.15, optionally 0.05 to 0.13,

Mn: up to 0.08, optionally 0.03 to 0.07,

Zn: up to 0.10, optionally up to 0.05,

impurities up to 0.05 each and up to 0.15 total, remainder aluminum,

(b) Machining said rolling ingot to a slab with a thickness of at least 10 in,

(c) Preheating said slab with minimum 2 hours soak at metal temperature of from 1000° F. to 1100° F. optionally from 1025° F. to 1075° F.,

(d) First hot rolling said slab with a break down mill to an intermediate plate with thickness of at least 1.1 in, optionally from 1.2 to 1.4 in,

(e) Second hot rolling said intermediate plate with a tandem mill, with an exit temperature from 650° F. to 690° F., optionally from 660° F. to 680° F. to a plate with a final thickness from 0.27 to 0.48 in, optionally from 0.31 to 0.43 in,

(f) Cutting said plate to obtain a flat blank for flow forming.

2. A method according to claim 1 wherein in (d), the entry temperature is at least 850° F. optionally at least 870° F. and/or is at most 950° F. optionally at most 930° F.

3. A flat blank for flow forming obtainable by the method of claim 1.

4. A flat blank according to claim 3 wherein the TYS is from 8 to 25 ksi in any direction and wherein the spread of TYS in any direction is at most 12 ksi.

5. A flat blank according to claim 3 wherein the UTS is from 25 to 30 ksi in any direction and wherein spread of UTS in any direction is at most 10 ksi.

6. A flat blank according to claim 3 wherein the grain size measured at mid-thickness on a L/ST section in the short direction is at least 40 μm , optionally at least 43 μm .

7. A flat blank according to claim 3 wherein the difference between hardening texture, quantified by the sum of the bulk volume fraction of “copper”, “brass” and “S” orientations and softening texture, quantified by a sum of a bulk volume fraction of “Cube”, “Goss” and “Rotated cube” orientations is no larger than 7%, optionally no larger 6%, optionally, no larger than 5%.

8. A flat blank according to claim 3 for flow forming.

9. A flat blank according to claim 8 for manufacturing by flow forming a cylindrical part that has side walls with a thickness of at least 0.08 in optionally at least 0.09 in and/or a cylindrical dimension with a length of at least 8 in optionally at least 9 in.

10. A flat blank according to claim 9 wherein the cylindrical part is used to make a transmission housing.

11. A flat blank with a thickness from 0.27 to 0.48 in, optionally from 0.31 to 0.43 in,

having a composition, in weight %

Mg: 2.2-2.6, optionally 2.3 to 2.5,

Cr: 0.16-0.25, optionally 0.19 to 0.22,

Fe: 0.10-0.35, optionally 0.20 to 0.30,

Cu: 0.01-0.09, optionally 0.02 to 0.07,

Ti: 0.005-0.15, optionally 0.01 to 0.05,

Si: up to 0.15, optionally 0.05 to 0.13,

Mn: up to 0.08, optionally 0.03 to 0.07,

Zn: up to 0.10, optionally up to 0.05,

impurities up to 0.05 each and up to 0.15 total, remainder

aluminum.

12. A flat blank according to claim 11 wherein the TYS is from 8 to 25 ksi in any direction and wherein the spread of TYS in any direction is at most 12 ksi.

13. A flat blank according to claim 11 wherein the UTS is from 25 to 30 ksi in any direction and wherein spread of UTS in any direction is at most 10 ksi.

14. A flat blank according to claim 11 wherein the grain size measured at mid-thickness on a L/ST section in the short direction is at least 40 μm , optionally at least 43 μm .

15. A flat blank according to claim 11 wherein the difference between hardening texture, quantified by the sum of the bulk volume fraction of “copper”, “brass” and “S” orientations and softening texture, quantified by a sum of a bulk volume fraction of “Cube”, “Goss” and “Rotated cube” orientations is no larger than 7%, optionally no larger 6%, optionally, no larger than 5%.

16. A flat blank according to claim 11 for flow forming.

17. A flat blank according to claim 11 for manufacturing by flow forming a cylindrical part that has side walls with a thickness of at least 0.08 in optionally at least 0.09 in and/or a cylindrical dimension with a length of at least 8 in optionally at least 9 in.

18. A flat blank according to claim 11 wherein the cylindrical part is used to make a transmission housing.

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